

Falaq Jad

AI Engineer | Data Science | ML | Computer Vision

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Profile

Final-year engineering student at ENSIAS, Artificial Intelligence Engineering program (2IA), graduating in June 2027. Convinced that an AI system only becomes meaningful once it moves beyond the notebook to reliably address a real-world need. Currently seeking a final-year graduation internship (PFE) in AI Engineering starting January 2027.

Education

École Nationale Supérieure d'Informatique et d'Analyse des Systèmes (ENSIAS) Engineering Degree, Artificial Intelligence Engineering program (2IA) Key courses: Deep Learning, Machine Learning, Computer Vision, Online Learning, Signal Processing, NLP, Intelligent Agents, MLOps, ML/AI Model Deployment, Databases and Big Data.	Sep 2025 – Jun 2027 Rabat, Morocco
CPGE Mohammed V Mathematics-Physics Track	Sep 2022 – Jun 2024 Casablanca, Morocco

Experience

Renault Digital Maroc AI Engineer Intern – Quality Engineering Team - Designed and developed a multi-agent AI system (3 agents) converting natural-language business requirements into Jira/GitLab tickets, Gherkin scenarios, and user stories, with human validation at each step. - Implemented a RAG pipeline with Ollama embeddings (nomic-embed-text), cosine similarity, and empirical calibration of the relevance threshold. Built a data anonymization layer (Presidio, spaCy) with 100% local LLM inference to meet the company's confidentiality requirements. - Automated (Copilot Studio / Power Automate) the weekly generation of tracking files on SharePoint, with comprehensive error handling and real-time Teams notifications; in production, used weekly by the QA team.	Jun 2026 – Aug 2026 Casablanca, Morocco
AKWA GROUP Data / Machine Learning Intern - Designed an end-to-end machine learning pipeline to detect and classify missing transactions, covering data preprocessing, feature engineering, modeling, and evaluation. - Tested and compared Logistic Regression, SVM, Decision Tree, Random Forest, and KNN models, with Random Forest achieving 98% overall accuracy and an 85% F1-score on missing transaction detection. - Collaborated with the Data team to automate the detection and ERP integration of missing transactions across more than 6 million records, gaining hands-on experience with large-scale data processing and real-world data quality challenges.	Jun 2025 – Sep 2025 Casablanca, Morocco

Projects

Dental Pathology Detection MLOps - Object detection model (YOLOv8s) identifying 7 categories of dental pathologies across 1,808 panoramic radiographs (24,888 annotations), prioritizing recall for clinical use (86% precision, 47% recall, 42% mAP50 on the test set). - Industrialized the model into a containerized API service with a complete MLOps pipeline (DVC, MLflow) and CI/CD, continuously deployed on Railway and Vercel.	🔗
Shoplifting Detection Computer Vision - Real-time detection system for suspicious behavior in surveillance videos; accuracy $\approx$ 80%. - Pipeline combining YOLO (detection), DeepSORT (tracking), MediaPipe Pose (33 landmarks), and an LSTM model (temporal behavior analysis).	🔗
Speaker Recognition Signal Processing & ML - Robust speaker identification system for degraded radio environments (SNR 10-15 dB), 86.98% accuracy across 50 speakers. - MFCC and spectral feature extraction (140 features); SVM classification (RBF kernel) with a real-time Streamlit interface.	🔗
Smart Irrigation System IoT & Machine Learning - Remote irrigation management platform (Flutter application, Firebase, ESP32/Arduino IoT architecture). - Machine learning model predicting optimal watering duration and timing from historical data (soil moisture, temperature), supporting farmer decision-making.	🔗

Technical Skills

Machine Learning & Deep Learning: PyTorch, Scikit-learn, XGBoost, YOLO, DeepSORT, MediaPipe, LSTM, SVM, CNN, GNN, Federated Learning, classification, anomaly detection
Generative AI & Agents: RAG, LangChain, LangGraph, NLP, LLM (Ollama), pretrained Transformer models
Data & Signal Processing: Python, NumPy, Pandas, SQL, MFCC, spectral feature extraction, feature engineering, time series, precision-recall evaluation
MLOps & Deployment: Docker, Docker Compose, MLflow, DVC, CI/CD, Airflow, FastAPI, REST APIs, GitHub Actions, Streamlit
Tools, Databases & IoT: Git, Big Data, PostgreSQL/Supabase, NoSQL, MongoDB, Flutter, Firebase, ESP32, Arduino
Languages: Arabic (Native), French (Fluent), English (Fluent)